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Model architecture dominates nutritional estimation accuracy in vision-language systems

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Vision-language models (VLMs) enable automated dietary assessment from food photographs, yet optimal deployment strategies remain unestablished. We evaluated 40 VLMs across eight providers using Nutrition5k database and original dishes pictures, testing image configurations (1–5 angles), prompt strategies (7 approaches), and image quality conditions, benchmarked against professional nutritionists. Model architecture dominated performance variance (99.6%). Multi-angle photography provided no benefit over single images (RMSLE: 0.627 vs. 0.623, $p = 0.182$), and prompt engineering showed no significant effects after correction (all $p > 0.05$). Contrary to conventional assumptions favoring standardized conditions, high-quality consumer smartphone photography significantly outperformed controlled laboratory imaging despite 56% smaller sample size (RMSLE: 0.548 vs. 0.616, $p = 0.020$, Cliff's $\delta = -0.58$). Ingredient descriptions slightly improved aggregate performance (0.548→0.516, $p = 0.039$) with heterogeneous model responses (−19.1% to +3.4%). Professional nutritionists substantially outperformed all VLMs (RMSLE: 0.176 vs. 0.443 best AI, 152% gap), particularly for protein estimation (672% worse AI performance). These results establish that model selection and image quality dominate nutritional estimation accuracy, while multi-view imaging and prompt complexity are negligible factors. Current VLMs suit consumer calorie tracking applications; achieving clinical-grade macronutrient profiling will require advances in model architecture rather than deployment optimization.

Keywords Nutritional estimation, Vision-language models, Computer vision, Dietary assessment, Deep learning, Food recognition, Prompt engineering, Automated nutrition analysis

Accurate dietary assessment remains a cornerstone of nutritional epidemiology, clinical nutrition, and public health research. Traditional methods—24-hour dietary recalls, food frequency questionnaires, and weighed food records—although considered gold standards, face well-documented limitations including recall bias, social desirability effects, and substantial participant burden. Notably, systematic underreporting of energy intake can reach 20–40% in obese populations[1,2], severely compromising both large-scale epidemiological research and the implementation of personalized dietary interventions. These limitations have historically constrained both the validity of research and the clinical applicability of dietary assessment tools.

The advent of large-scale vision-language models (VLMs) has opened unprecedented opportunities for automating dietary assessment. These systems integrate deep visual processing through Vision Transformers with contextual linguistic understanding derived from Large Language Models (LLMs), demonstrating multimodal reasoning capabilities that surpass traditional computer vision systems[3,4]. Contemporary models such as GPT-4 Vision, Gemini, and Claude can not only identify foods in photographs but also infer preparation methods, estimate portions through visual cues, and generate nutritional breakdowns through multi-step reasoning processes[5,6]. In parallel, hybrid approaches combining Convolutional Neural Networks for local texture extraction with Transformers for global dependency modeling have achieved accuracies exceeding 99% in controlled food classification tasks[7].

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However, the transition from impressive performance in controlled research settings to reliable deployment in real-world applications remains problematic. Existing validation studies reveal that even advanced VLMs achieve accuracies of 65–70% on comprehensive food recognition tasks and exhibit systematic underestimations of 20–50% for caloric intake and macronutrients, with particular difficulty in quantifying invisible ingredients such as cooking oils and condiments[5,6,8]. Moreover, the literature lacks systematic comparisons between prompt engineering strategies specifically for nutritional estimation tasks, and the impact of sampling parameters (temperature, top-p) on vision task accuracy remains understudied[9].

Beyond model selection, two practical deployment questions emerge for image-based nutritional assessment. First, does capturing food from multiple angles—analogous to human visual inspection of a meal—improve estimation accuracy? Depth and volume estimation from single 2D images is an inherently ill-posed problem, suggesting that stereo or multi-view imaging might reduce uncertainty. Second, can prompt engineering techniques that have proven successful in text-based reasoning tasks (chain-of-thought, few-shot learning, expert role-playing) transfer to vision-language nutritional estimation? If complex prompts guide models through a systematic assessment of portion size and ingredient identification, accuracy might improve.

The present study addresses these questions through systematic experimentation on the Nutrition5k dataset, a benchmark comprising food dishes with ground-truth nutritional values. We evaluated 40 VLMs across multiple experimental conditions to determine: (1) whether multi-image configurations (2, 3, or 5 images) improve upon single-image estimation; (2) whether advanced prompt engineering strategies enhance accuracy; and (3) which factors—model architecture, temperature, image quality, or dataset composition—most strongly influence performance.

Our findings have implications for both research and practical applications. If multi-view imaging confers no advantage, dietary assessment applications can avoid the complexity and user burden of capturing multiple photographs per meal. If prompt engineering proves ineffective or counterproductive, deployment can prioritize simple interfaces over elaborate instructions. Most fundamentally, understanding the limits of current VLMs for nutritional estimation will inform whether specialized training, fine-tuning, or hybrid approaches combining VLMs with nutritional databases are necessary for clinical-grade dietary assessment.

Materials and methods

Dataset

We used the Nutrition5k dataset, a publicly available benchmark developed by Google Research for automated nutritional analysis[10]. The dataset contains 5,006 food dishes photographed in a controlled laboratory setting, with ground-truth nutritional values. Each dish includes one RGB overhead image (high-quality, well-lit, centered) and stereo video recordings from two side-angle cameras. Although Nutrition5k also provides dense depth data from the 3D scanning rig, these were used only to derive the ground-truth nutritional values and were not provided to the evaluated models, whose input was restricted to RGB images (plus, where indicated, textual ingredient descriptions).

From this dataset, we selected 50 dishes stratified by calorie content across five ranges (49–100, 101–299, 300–499, 500–699, 700–999 kcal, with 10 dishes per range) to ensure representation across the nutritional spectrum. Selection was performed using random sampling with fixed seed values (123, 321, 995) to ensure reproducibility while testing robustness to dataset composition. We extracted the following nutritional variables for analysis: total energy (kcal), carbohydrates (g), proteins (g), and fat (g). Dietary fiber was excluded due to high rates of missing data in the ground-truth measurements.

For multi-image experiments, we downloaded stereo videos (camera A and camera B, h264 format) and extracted frames using FFmpeg (version 4.4.2). From each camera (that rotates around the plate of ~ 180 degrees from the first to the last frame of the video), we extracted the first and last usable frames, providing pictures from different angles of the same plate. This yielded four stereo images per dish (2 cameras × 2 time points), which were combined with the overhead RGB image for the 5-image configuration.

For high quality (HQ) images, a nutritionist (SP) provided the pictures of 23 dishes (single picture, from above) done with a commercial smartphone (Apple iPhone 16) and sent through an instant message application (Meta WhatsApp) to compress them and simulate a real-life scenario. High quality images were used to evaluate LLM performance both as-is (HQ setup) or with an-image specific prompt describing the food content (but not the quantity) of the picture (HQ+Desc setup).

Every picture's nutritional values were evaluated by the same nutritionist in two ways: an "at glance" evaluation (i.e., images were evaluated according to the nutritionist experience) simulating a similar process an LLM will use for its evaluation, and a systematic evaluation using precise weights and standard Italian nutritional tables provided by the National Research Institute for Food and Nutrition (INRAN). These two evaluations were used as "human benchmark" to be compared to the LLM results. The study was approved by IRCCS Istituto Oncologico "Gabriella Serio", ID: 1619/CEL of 10.06.2024. All methods were carried out in accordance with relevant guidelines and regulations. No informed consent was required since no human subject was included in the study.

Vision-language models

We evaluated 40 vision-language models accessed via the OpenRouter API (October 2025 versions), spanning eight major providers:

- OpenAI (9 models): GPT-5 series (gpt-5, gpt-5-mini, gpt-5-nano, gpt-5-chat), GPT-4.1 series (gpt-4.1, gpt-4.1-mini, gpt-4.1-nano), GPT-4o series (gpt-4o, gpt-4o-mini).
- Anthropic (7 models): Claude Opus 4 series (claude-opus-4, claude-opus-4.1), Claude Sonnet 4 series (claude-sonnet-4, claude-sonnet-4.5), Claude 3.7 Sonnet, Claude 3 Opus, Claude 3 Haiku.

- Google (6 models): Gemini 2.5 series (gemini-2.5-flash, gemini-2.5-flash-lite, gemini-2.5-pro), Gemini 2.0 (gemini-2.0-flash-001), Gemma 3 series (gemma-3-27b-it, gemma-3-12b-it).
- Mistral AI (4 models): Pixtral series (pixtral-12b, pixtral-large-2411), Mistral Small, Mistral Medium.
- Other (11 models): Qwen Vision series (qwen3-vl-235b, qwen2.5-vl-32b-instruct), Meta Llama Vision (llama-3.2-90b-vision-instruct, llama-4-scout, llama-4-maverick), X.AI Grok (grok-4, grok-4-fast), ByteDance UI-TARS 1.5-7b.
- Additional models (3 models): these models were evaluated in a single image configuration setting because they became available after the other settings were tested. OpenAI gpt-5-pro, Anthropic claude-haiku-4.5, Openrouter andromeda-alpha.

The single-image performance of the three Additional models, together with that of three further models released between the initial submission of this manuscript and its revision (Google gemini-3-pro-preview, OpenAI gpt-5.1, OpenAI gpt-5.1-chat), is reported in Supplementary Table S2.

The evaluation followed a two-stage design. All models were first screened on the single-image configuration under standardized conditions, producing a primary performance ranking. From this screening we selected a subset of nine models for the more computationally demanding deep evaluation (multi-image configurations, prompt-engineering strategies, and ingredient-description conditions) to keep the combinatorial API cost tractable while retaining architectural diversity. The subset was chosen to include the best-ranked model from each of the four major providers and to span the full baseline-RMSLE range of the screened ensemble, so that the effects of deployment axes could be observed across architectures of different baseline quality rather than only across the top performers.

Models were selected to represent diverse architectural approaches, parameter scales (12B to 235B parameters), and cost profiles (premium commercial to free open-source). Models were evaluated on single-image configuration (see below) to test specific image characteristics and temperature influence.

Image configurations

We tested four configurations: (1) 1-image: Single RGB overhead photograph (2) 2-image (stereo): Two side-angle stereo frames (camera A first frame + camera B first frame) (3) 3-image (stereo + RGB): Overhead RGB + two stereo frames (4) 5-image (complete): Overhead RGB + four stereo frames (2 cameras × 2 time points).

Experimental Parameters

To assess robustness, we varied: - Temperature: 0.0 (deterministic), 0.1 (default), 0.3 (moderate variability) - Random seed: 123, 321, 995 (controlling dish selection for independent test sets) - Maximum tokens: 3,600 (standard models).

This factorial design yielded 5 seed×temperature configurations for single-image analysis (3 seeds at temperature 0.1, plus seed 123 at temperatures 0.0 and 0.3), totaling 9,050 predictions for initial model screening.

Prompt engineering strategies

We evaluated seven prompting approaches on a subset of 9 models across 50 dishes: (1) Baseline: "Analyze this food image and estimate nutritional content for the entire portion. Provide precise numbers." (2) Chain-of-Thought: Explicit step-by-step reasoning (identify food items → estimate portion size → calculate per-item nutrition → sum total) (3) Expert Role: Frame the model as a nutritionist with 20 years of experience, with explicit error-prevention guidelines (4) Comparative Anchoring: Provide reference portion sizes with known nutritional values for calibration (5) Uncertainty-Aware: Acknowledge limitations in hidden ingredients and cooking methods, then provide best estimate (6) Multi-Step with Self-Correction: Iterative reasoning with sanity checks and correction factors for hidden oils/fats (7) Few-Shot Learning: Three worked examples before the target image. Exact prompt used are available in the linked code repository.

API implementation

All models were accessed via the OpenRouter unified API using the *ellmer* R package (version 0.3.2) for structured output and standard JSON-based inference. We implemented a hybrid approach:

1. Structured output (preferred): For models supporting native JSON schema enforcement (GPT-4/5, Gemini), we used structured completions with predefined nutritional fields.
2. JSON fallback: For models without structured output support (Claude, Mistral, Qwen, Llama), we used standard chat completion with manual JSON parsing from response text.

Images were preprocessed using the *magick* R package, resized to high quality for API transmission, and encoded as base64 where necessary. To avoid rate limiting, we introduced 5-second delays between sequential API calls and an automatic retry for unresponsive models.

Performance metrics

The selected primary metric was Root Mean Squared Logarithmic Error (RMSLE). RMSLE was chosen as the primary metric due to its robustness to heterogeneous scales (kcal vs. grams) and its emphasis on relative rather than absolute errors:

$$\mathrm{RMSLE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left[\log\left(\mathrm{pred}_i + 1\right) - \log\left(\mathrm{actual}_i + 1\right)\right]^2}$$

RMSLE penalizes both underestimation and overestimation proportionally, handles zero values gracefully, and aligns with established nutritional estimation benchmarks.

Secondary metrics were Symmetric Mean Absolute Percentage Error (sMAPE) (bounded between 0 and 200%, symmetric treatment of over/underestimation); Mean Absolute Error (MAE) (interpretable in original units kcal, grams for clinical relevance); Interquartile Range (IQR) (measures prediction consistency using Median Absolute Deviation, MAD, normalization). We used macro-averaging across the four nutrients (energy, carbohydrates, proteins, fat), giving equal weight to each nutrient:

Overall_RMSLE = (RMSLE_kcal + RMSLE_carbs + RMSLE_proteins + RMSLE_fat) / 4.

Statistical analysis

Performance differences between experimental conditions were assessed using non-parametric tests appropriate for repeated measures and non-normal distributions. Friedman test assessed differences across four image configurations with models as blocking factor; Kruskal-Wallis test compared independent groups (e.g., prompt strategies); Wilcoxon signed-rank test tested pairwise comparisons with Bonferroni correction; Cliff's Delta estimated the effect size (negligible: $|\delta| < 0.147$, small: 0.147–0.330, medium: 0.330–0.474, large: ≥0.474). Variance decomposition for experimental factors (seed, temperature, model) was performed using a descriptive variance-component analysis based on the variance of factor-level group means, relative to the total variance of *Overall_RMSLE*. For each factor, between-group variance was calculated by: (1) computing group means while holding other factors constant (e.g., seed variance calculated at temperature = 0.1; temperature variance calculated at seed = 123), and (2) calculating the variance of these group means. Within-model variance was computed as the mean of within-configuration variances across all seed × temperature combinations. The percentage of total variance attributable to each factor was calculated as (factor variance / total variance) × 100. All analyses were conducted in R (version 4.5.1)[11].

Experimental workflow

Figure 1. Experimental workflow. The Nutrition5k dataset was stratified into 5 kcal groups (49–999 kcal, 10 dishes per group). We evaluated 40 vision-language models (VLMs) across three experimental dimensions: dataset seed (3 random samples), temperature (0.0, 0.1, 0.3), and image configurations (1, 2, 3, 5 images per dish). From initial screening, 9 best-performing models were selected for extensive prompt engineering evaluation (7 strategies: baseline, chain-of-thought, expert role, anchoring, uncertainty-aware, few-shot, multi-step). Finally, best performing models were tested also on high-resolution original images. Created in BioRender. Vedovelli, L. (2026) https://BioRender.com/a6lvrz9.

Results

Experimental parameters (seed and temperature) effects

Within seed 123, temperature variation had negligible effects. Mean RMSLE ranged from 0.687 (T = 0.1) to 0.692 (T = 0.3), a 0.7% difference. Pairwise comparisons showed no significant differences (all adjusted $p > 0.05$, Cliff's $\delta < 0.04$).

Across three independent 50-dish samples (seeds 123, 321, 995), performance varied by 1.4% (mean RMSLE 0.680–0.690), reflecting inherent difficulty variation in dish composition. Kruskal-Wallis test showed no significant seed effect ($H = 2.34$, $p = 0.31$).

Multi-image configuration analysis

Based on performance, architecture, and cost efficiency, we selected 9 top-performing models for multi-image evaluation (Table 1).

Analysis of 1,765 predictions (9 models × 50 dishes × 4 configurations) revealed no statistically significant performance improvement from using multiple images (Table 2). Mean RMSLE varied minimally across configurations: 0.627 (1-image), 0.639 (2-image stereo), 0.618 (3-image stereo + RGB), 0.623 (5-image complete). Friedman test indicated no significant differences ($\chi^2(3) = 4.867$, $p = 0.182$).

When the same data are reorganized by strategy rather than by model (Supplementary Table S1), the identity of the top-performing model is almost invariant to the choice of strategy (OpenAI GPT-5-chat leads on 5 of 7 strategies and OpenAI GPT-4.1 on the remaining 2) and the spread of the best-achievable RMSLE across strategies is only 0.030, further reinforcing that model choice dominates prompt choice.

Pairwise Wilcoxon tests detected no evidence for a difference between any configuration pairs (all adjusted $p > 0.05$). Effect sizes were negligible to small (Cliff's δ range: -0.11 to +0.21). The 2-image stereo configuration showed slightly higher mean RMSLE and IQR, suggesting increased rather than decreased prediction variability.

All four macronutrients exhibited similar insensitivity to image count. RMSLE variation across configurations ranged from 0.01 to 0.03 for all nutrients. Proteins showed the most consistent performance (RMSLE 0.57–0.60), while energy showed the highest absolute errors (RMSLE 0.67–0.69) across all configurations.

Model rankings remained highly stable across configurations (Spearman $\rho = 0.94$–0.98 for all pairwise comparisons). The top performer, OpenAI GPT-5-chat, achieved RMSLE of 0.545 (1-image), 0.550 (2-image), 0.514 (3-image), and 0.514 (5-image). No model showed selective advantage from multi-image input.

Multi-image configurations incurred proportional API cost increases (2×, 3×, 5× for respective setups) without corresponding performance gains. The cost-benefit ratio was consistently negative across all image counts.

Figure 2. Model ranking stability heatmap. RMSLE values for 9 models across 4 image configurations. Models are ordered by mean performance across all configurations. Color intensity indicates RMSLE magnitude (darker = better). High correlation between columns (Spearman $\rho > 0.94$) confirms stable rankings.

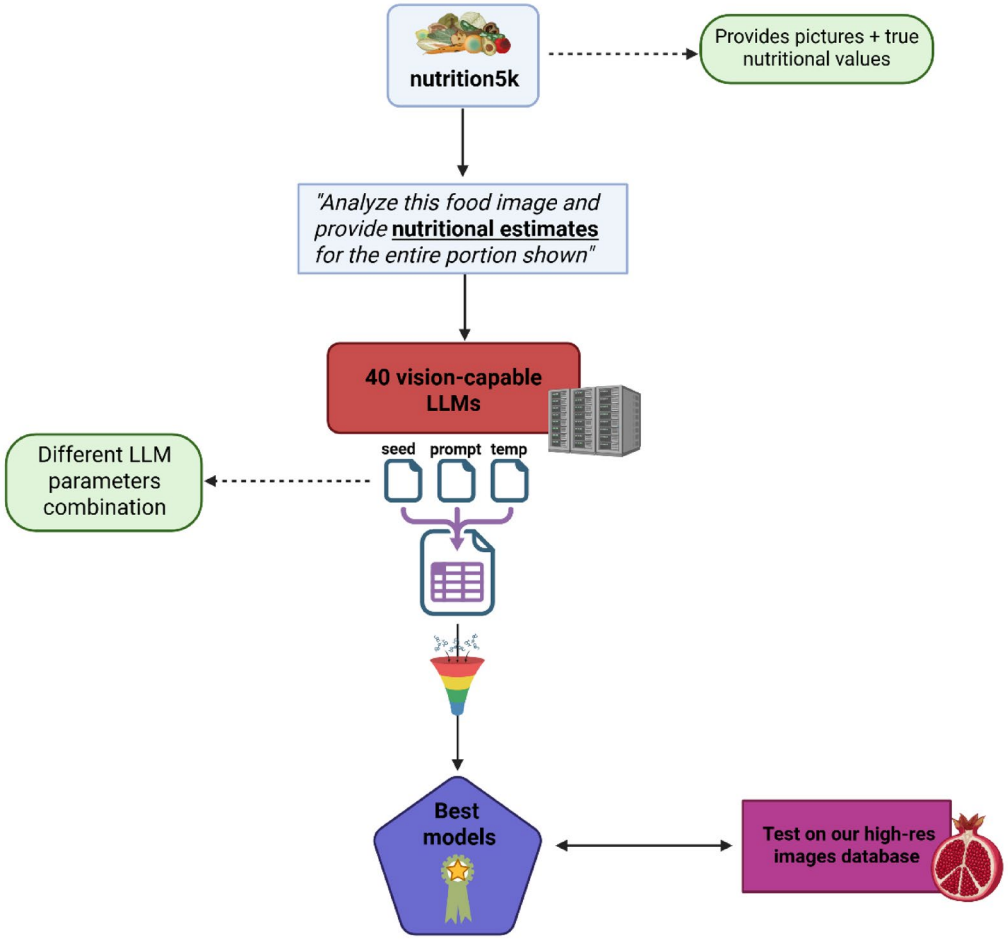


Fig. 1. illustrates the complete experimental pipeline, from dataset sampling through model evaluation to final validation on mobile applications and high-resolution original images.

Model	Reason
Anthropic/claude-3.7-sonnet	Anthropic best small
Anthropic/claude-opus-4	Anthropic best overall
Google/gemini-2.0-flash	Quick and cheap
Google/gemini-2.5-flash	Google best overall
Mistralai/pixtral-12b	Best opensource
Openai/gpt-4.1	Consistent performance (API only)
Openai/gpt-4o	Proved workhorse
Openai/gpt-5-chat	Best overall
Qwen/qwen2.5-vl-32b-instruct	Best free

Table 1. Selected models for downstream analysis.

Configuration	*N* Images	Mean RMSLE	Median RMSLE	Mean sMAPE (%)	Mean IQR
1-image	1	0.627	0.614	51.1	0.974
2-image stereo	2	0.639	0.643	51.5	1.082
3-image stereo + RGB	3	0.618	0.616	50.0	0.993
5-image complete	5	0.623	0.621	49.8	1.018

Table 2. Performance metrics by image configuration.

Rank	Strategy	Mean RMSLE	Δ vs. Baseline	*p*-value
1	Baseline (simple)	0.633	—	—
2	Uncertainty-Aware	0.628	-0.8%	1.000
3	Expert Role	0.651	+2.9%	1.000
4	Chain-of-Thought	0.656	+3.7%	1.000
5	Few-Shot Learning	0.666	+5.2%	0.328
6	Comparative Anchoring	0.666	+5.3%	0.445
7	Multi-Step Reasoning	0.684	+8.0%	0.117

Table 3. Performance by prompt engineering strategy. *p-values from Wilcoxon signed-rank tests vs. baseline, Bonferroni-corrected.*

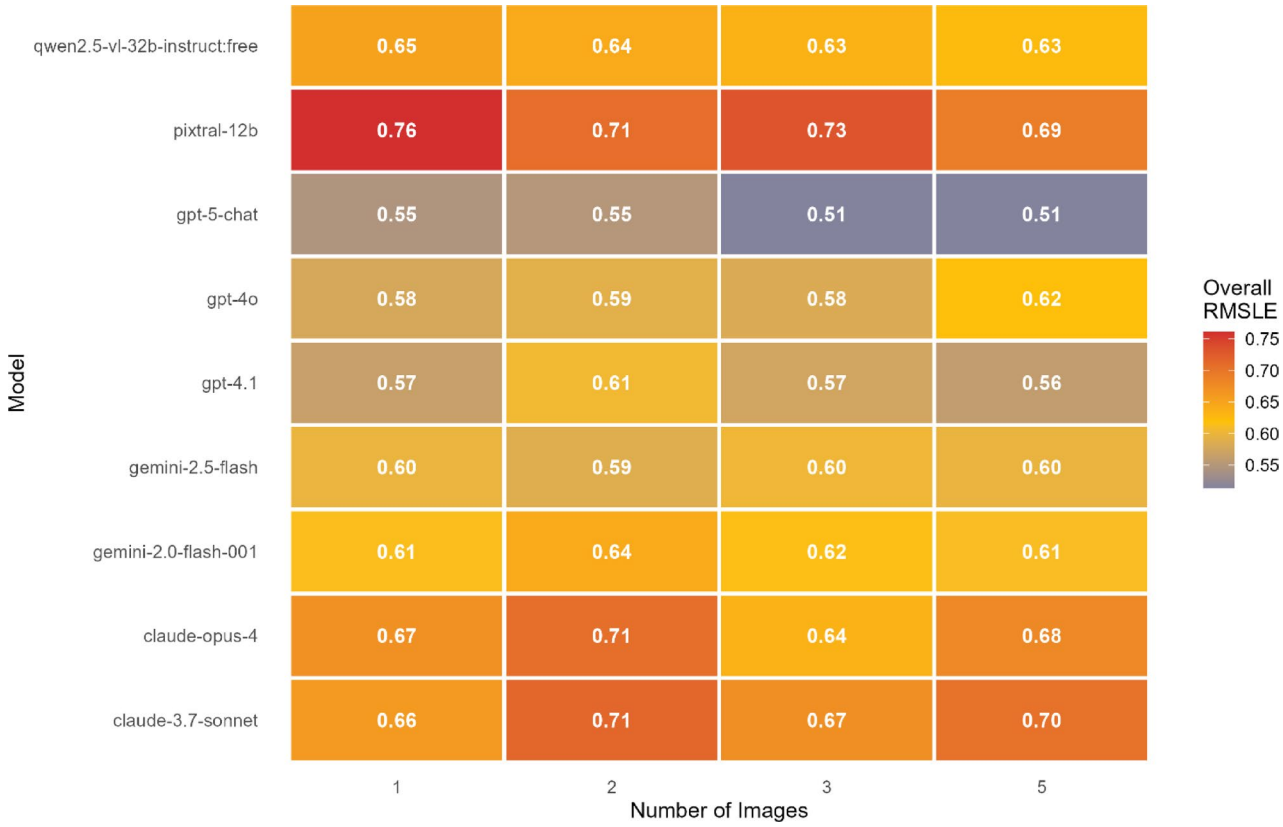


Fig. 2. illustrates model ranking stability through a heatmap of performance across all configurations, confirming that relative model performance remained consistent regardless of image count.

Prompt engineering evaluation

Analysis of 3,150 predictions (7 strategies × 9 models × 50 dishes) revealed that simple baseline prompts equaled or outperformed all advanced prompting strategies (Table 3). The baseline achieved mean RMSLE = 0.633, statistically tied with uncertainty-aware prompting (0.628), while all other strategies degraded performance.

Multi-step reasoning with self-correction showed the largest numerical difference from baseline (+8.0% RMSLE), though this did not reach statistical significance ($p = 0.117$). Chain-of-thought prompting, proven effective in mathematical reasoning tasks, similarly showed no significant difference from baseline ($p = 1.000$). The uncertainty-aware strategy performed marginally better than baseline (−0.8%), but again without statistical significance ($p = 1.000$).

Strategies' effects on single models

While aggregate analysis revealed no statistically significant differences between prompt strategies (Table 3), examination of individual model responses showed heterogeneous effects. Table 4 presents the best-performing strategy for each model and the performance delta relative to its baseline.

Three models showed meaningful improvements (>4%) with specific strategies: Claude 3.7 Sonnet achieved an 8.4% RMSLE reduction with uncertainty-aware prompting (0.693 → 0.635), GPT-4.1 improved 4.7% with expert role prompting (0.559 → 0.533), and GPT-5-chat gained 3.8% from comparative anchoring (0.549 →

Model	Best Strategy	RMSLE	Δvs Baseline	Baseline RMSLE
OpenAI GPT-5-chat	Comparative Anchoring	0.528	−3.8%	0.549
OpenAI GPT-4.1	Expert Role	0.533	−4.7%	0.559
OpenAI GPT-4o	Uncertainty-Aware	0.592	−0.6%	0.595
Anthropic Claude 3.7 Sonnet	Uncertainty-Aware	0.635	−8.4%	0.693
Anthropic Claude Opus-4	Baseline	0.679	0.0%	0.679
Google Gemini 2.5 Flash	Uncertainty-Aware	0.573	−3.1%	0.591
Google Gemini 2.0 Flash	Chain-of-Thought	0.602	−1.0%	0.609
Qwen 2.5-VL-32B	Expert Role	0.643	−2.0%	0.656
Mistral Pixtral-12B	Uncertainty-Aware	0.760	−0.6%	0.764

Table 4. Best strategy per model.

0.528). Conversely, Claude Opus-4 performed best with the simple baseline prompt, and Pixtral-12B showed negligible variation across strategies (< 0.6% range).

The uncertainty-aware strategy emerged as the most frequently optimal approach (4 of 9 models), followed by expert role (2 models), and single instances of comparative anchoring, chain-of-thought, and baseline. This suggests that explicit acknowledgment of estimation uncertainty may provide modest benefits for certain model architectures, particularly Anthropic's Claude 3.7 and Google's Gemini 2.5.

However, the lack of statistical significance in this (all $p > 0.05$) and on the aggregate analysis (Table 3) indicates these observed improvements are either: (1) too small relative to between-dish variance to be reliably detected, (2) model-specific effects that average out across the ensemble, or (3) potential sampling artifacts from the 50-dish evaluation set. The largest individual improvement (Claude 3.7: −8.4%) corresponds to an absolute change in RMSLE of only 0.058, which remains within the typical measurement uncertainty for visual nutritional estimation tasks.

The sign and magnitude of prompt-induced change varied substantially across architectures and did not follow a clean monotonic relationship with baseline RMSLE. The largest deteriorations under reasoning-heavy strategies were observed for Claude Opus-4 (chain-of-thought: 0.679 → 0.865, +27%; multi-step: +24%) and Pixtral-12B (multi-step: 0.764 → 0.878, +15%), whereas Claude 3.7 Sonnet, despite a comparable baseline to Claude Opus-4, improved by 8.4% under uncertainty-aware prompting, indicating that prompt responsiveness is not captured by baseline accuracy alone. A tentative interpretation is that strategies requiring extended multi-step inference (chain-of-thought, multi-step self-correction, expert role) only help architectures that can support that inference, while metacognitive strategies that reframe the task without imposing intermediate reasoning steps (uncertainty-aware, comparative anchoring) tend to produce smaller but more uniformly non-negative effects. Given the modest number of models and strategies, this mechanistic attribution remains exploratory.

Overall contribution of different setups and parameters

Overall, quantified variance contributions were model architecture (99.6%), dataset seed (0.3%), temperature (0.1%). Model selection dominated all other factors by a factor of 2 orders of magnitude.

Best performing models

The top 5 models on single-image configuration (seed 123, temperature 0.1):

1. OpenAI GPT-5-chat: RMSLE 0.545, sMAPE 45.6%, MAE 37.1 kcal.
2. OpenAI GPT-4o: RMSLE 0.614, sMAPE 48.9%, MAE 42.3 kcal.
3. OpenAI GPT-4.1: RMSLE 0.621, sMAPE 49.2%, MAE 43.7 kcal.
4. Google Gemini 2.5 Flash: RMSLE 0.647, sMAPE 50.8%, MAE 45.9 kcal.
5. Google Gemini 2.0 Flash: RMSLE 0.661, sMAPE 51.4%, MAE 47.2 kcal.

The best model (GPT-5-chat) achieved mean absolute errors of approximately 37 kcal for energy, 8.5 g for carbohydrates, 6.2 g for proteins, and 5.8 g for fat.

Four-way comparison: human benchmark, image quality, and textual context

To systematically evaluate the factors influencing nutritional estimation accuracy, we conducted a comprehensive four-way comparison across different expertise sources (human vs. AI), image quality levels, and information modalities.

High-quality Images (HQ) - best AI performance

On high quality (HQ) images, openai/gpt-5-chat achieved RMSLE of 0.443, representing a 152% increase over human expert performance (0.176). Openai/gpt-5-chat, ranked second with RMSLE of 0.498, outperforming all the other models (Table 5).

OpenAI's GPT-5-chat emerged as the clear leader with an RMSLE of 0.443, outperforming the second-place GPT-4o by 12.4%. The performance spread between the best and worst models was 1.35×, indicating moderate but consistent architectural differences in vision-language nutritional estimation capabilities. Notably,

Rank	Model	RMSLE ↓	sMAPE (%) ↓	MAE (%) ↓	vs. Human Gap
1	openai/gpt-5-chat	0.443	29.2	23.3	+ 152%
2	openai/gpt-4o	0.498	35.1	25.7	+ 183%
3	google/gemini-2.0-flash-001	0.527	38.8	29.0	+ 200%
4	mistralai/pixtral-12b	0.555	40.6	38.9	+ 215%
5	google/gemini-2.5-flash	0.564	40.5	35.0	+ 220%

Table 5. Model performance ranking:

Model	HQ	HQ+Desc	Nutrition5k	Best Setup	Spread
openai/gpt-5-chat	0.443	0.440	0.528	HQ+Desc	1.20×
openai/gpt-4.1	0.583	0.569	0.533	Nutrition5k*	1.09×
anthropic/claude-3.7-sonnet	0.580	0.469	0.635	HQ+Desc	1.35×
mistralai/pixtral-12b	0.555	0.528	0.760	HQ+Desc	1.44×

Table 6. Cross-Setup Model Performance (Selected Models). *Note: GPT-4.1's best performance in Nutrition5k likely reflects strategy optimization rather than inherent suitability, as aggregate analysis across all models confirms HQ superiority.

Category*	Mean RMSLE	Range RMSLE	Characteristics
Easy tier (RMSLE < 0.4)	0.283	0.212–0.397	Mostly single-component, homogeneous, visible ingredients
Moderate tier (0.4–0.8)	0.590	0.403–0.792	Multi-component dishes, some hidden elements
Difficult tier (> 0.8)	1.058	0.843–1.367	Multi-component, hidden fats, complex textures
Overall	0.612	0.212–1.367	546% max/min ratio

Table 7. Dish-level error distribution and tier classification. *Easy tier: kcal = 0.196, carbs = 0.257, protein = 0.215, fat = 0.220; Moderate tier: kcal = 0.520, carbs = 0.482, protein = 0.605, fat = 0.753; Difficult tier: kcal = 1.066, carbs = 0.706, protein = 0.995, fat = 1.855; Fat ratio (difficult / easy): 8.4× (primary error driver).

while all models substantially trailed human performance, the high-quality imagery enabled markedly better AI performance than standardized laboratory photography.

High-quality with descriptions (HQ+Desc) - multimodal configuration
We used 16 images paired with textual ingredient lists without quantitative information on 9 models (the same used to test prompting strategies). Example descriptions were:

- "pollo zucca delica" (chicken, delica pumpkin).
- "salmone selvaggio surgelato e piselli surgelati" (wild frozen salmon and frozen peas).

Statistical analysis across 9 common models revealed that ingredient descriptions significantly improved aggregate performance compared to HQ visual-only baseline (mean RMSLE: 0.548 → 0.516, − 5.5% improvement, Wilcoxon signed-rank test: $V = 40.0$, $p = 0.039$, Cliff's $\delta = 0.481$, large effect). However, this aggregate benefit masked substantial heterogeneity in individual model responses. The single best-performing model (GPT-5-chat) showed minimal improvement (0.443 → 0.440, − 0.7%), while others demonstrated more pronounced benefits (e.g., Claude-3.7-sonnet: −19.1%, Qwen-2.5-VL: −8.9%). Two models even showed slight performance degradation (GPT-4o: +3.4%, Gemini-2.0-flash: +1.0%).

Cross-setup model consistency
Model consistency throughout setups varied considerably, with performance spread ranging from 1.09× (GPT-4.1, highly stable) to 1.44× (Pixtral-12B, highly sensitive to setup). Seven of nine models achieved their best performance in HQ+Desc, with HQ as second-best for most. Nutrition5k was consistently the poorest-performing setup, with only one model (GPT-4.1) achieving its best score in this configuration.

The Anthropic Claude family demonstrated the strongest multimodal integration capabilities, with Claude-3.7-sonnet achieving a 19.1% improvement when ingredient descriptions were added. This substantial gain suggests architectural innovations in cross-modal attention mechanisms that other model families have not yet replicated. Conversely, Google's Gemini-2.5-flash exhibited remarkable stability (1.04× spread), maintaining consistent performance regardless of setup—a desirable property for production deployment where input conditions may vary See table 6 and 7.

All three pairwise comparisons reached statistical significance, with large effect sizes indicating that the observed performance differences are robust and unlikely to be attributable to chance. The strongest effect

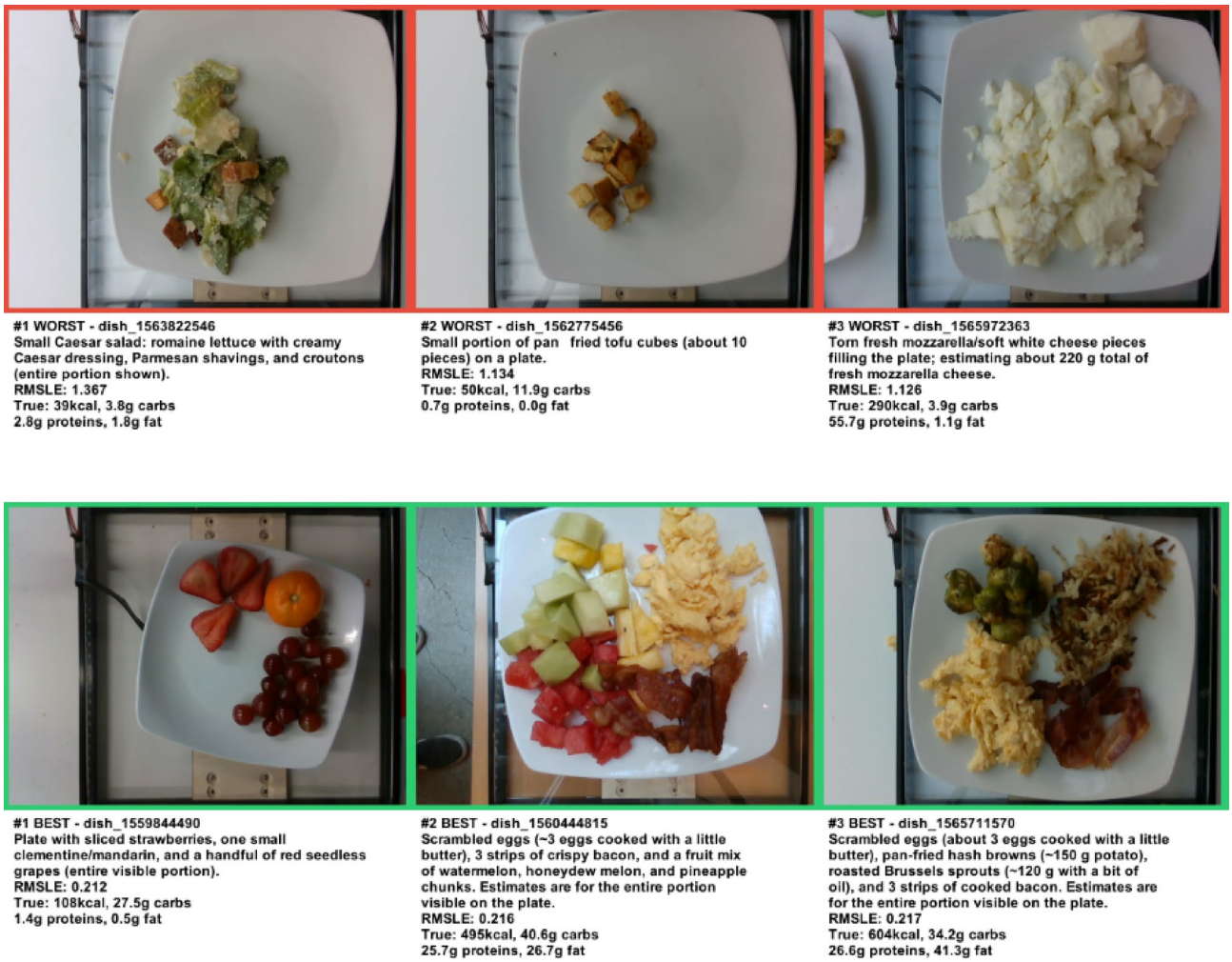


Fig. 3. Worst vs. best performing dishes. Visual comparison of the 3 most difficult (top, mean RMSLE = 1.166) and 3 easiest (bottom, mean RMSLE = 0.218) dishes. Difficult dishes show multi-component complexity, difficult textures, and hidden ingredients, while easy dishes exhibit more recognizable and unambiguous visual characteristics. For example, Mozzarella cheese (#3 worst) was often recognized as "egg-white".

emerged in the comparison between HQ+Desc and Nutrition5k ($p = 0.012$, $\delta = -0.85$), showing that high-quality consumer photography enriched with textual context yields substantially and consistently better results than standardized laboratory imaging.

The comparison between HQ and HQ+Desc, although statistically significant ($p = 0.039$), displayed the smallest effect size among the three ($\delta = 0.48$). This pattern reflects heterogeneous treatment effects, where aggregate improvements are muted by the minimal response of the single best-performing model. The positive Cliff's delta indicates that a greater proportion of pairwise model differences favor HQ+Desc over HQ, confirming directional consistency despite the variability in magnitude.

The contrast between HQ and Nutrition5k demonstrated a clear degradation in performance for the latter ($p = 0.020$, $\delta = -0.58$), while the HQ+Desc versus Nutrition5k comparison showed the largest relative improvement associated with enriched photographic inputs. Collectively, these findings underscore that both image quality and contextual textual information materially influence model behavior, with the combination of the two producing the most pronounced advantages.

Dish-level error analysis

Mean RMSLE ranged from 0.212 to 1.367 (546% difference), with a right-skewed distribution (median = 0.551, Fig. 1). We classified dishes into three difficulty tiers: easy (RMSLE < 0.4), moderate (0.4–0.8), and difficult (> 0.8). Food items associated with lower error values are typically those with more recognizable and unambiguous visual characteristics, achieving balanced performance across nutrients (RMSLE 0.196–0.257). Difficult dishes showed multi-component complexity and hidden ingredients, with dramatically elevated fat errors (mean fat RMSLE = 1.855 vs. 0.220 for easy dishes, 8.4× ratio). The worst-performing dish (dish_1563822546, RMSLE = 1.367) exhibited the highest fat estimation errors (RMSLE = 1.741), suggesting substantial hidden lipid content.

Rank	Setup	Performer	RMSLE ↓	sMAPE (%) ↓	MAE ↓	*N* Dishes
1st	Human	Professional Nutritionist	0.176	13.0	13.7	7
2nd	HQ+Desc	openai/gpt-5-chat	0.440	26.8	22.6	16
3rd	HQ	openai/gpt-5-chat	0.443	29.2	23.3	23
4th	Nutrition5k	openai/gpt-5-chat	0.528	46.8	35.6	50

Table 6. Four-way performance comparison. HQ: high-quality images, HQ+Desc: high-quality images + description of the image in the prompt.

Standard deviation of RMSLE across models ranged from 0.114 to 0.519, indicating that difficult dishes consistently challenge all architectures rather than showing model-specific failures. The worst-performing dish (dish_1563822546, RMSLE = 1.367) exhibited catastrophic fat estimation errors (RMSLE = 1.741) alongside elevated energy errors (1.738), suggesting substantial hidden lipid content from cooking oils or fatty meat preparations.

Fat estimation emerged as the primary driver of dish-level difficulty (Table 1). Difficult tier dishes showed 8.4× higher fat RMSLE than easy dishes (1.855 vs. 0.220), while carbohydrates exhibited the smallest gradient (2.7× ratio, 0.706 vs. 0.257). Protein estimation showed intermediate difficulty (4.6× ratio), and energy errors (6.0× ratio) correlated strongly with fat underestimation.

Decomposition of errors by macronutrient revealed that fat estimation is the primary driver of dish-level difficulty (Table 1). Across the difficult tier ($n = 10$ dishes, RMSLE > 0.8), mean fat RMSLE reached 1.855—8.4× higher than the easy tier (0.220). In contrast, carbohydrate estimation showed the smallest difficulty gradient (mean difficult tier = 0.706 vs. easy tier = 0.257, ratio 2.7×), indicating that carbohydrate-rich foods (rice, pasta, bread) maintain visual predictability even in complex preparations.

Protein estimation exhibited intermediate difficulty (4.6× ratio, difficult tier RMSLE = 0.995 vs. easy tier = 0.215), suggesting that while simple lean proteins are reliably estimated, mixed dishes with multiple protein sources or processed meats introduce substantial ambiguity. Energy estimation (6.0× ratio) correlated strongly with fat errors, confirming that total calorie inaccuracies are driven primarily by hidden lipid content rather than portion size misjudgment.

Overall performance hierarchy

Table 6 presents the best-performing configuration across all four experimental setups, establishing the human benchmark and AI capabilities under different conditions.

Professional human nutritionists substantially outperformed all AI systems, achieving an RMSLE of 0.176 compared to 0.440 for the best AI configuration (151.9% performance gap). However, formal statistical testing of this difference was not feasible due to different dish sets (7 human-assessed dishes vs. 16–50 AI-assessed dishes), rendering this a descriptive rather than inferential comparison.

Among AI configurations, high-quality consumer photography with ingredient descriptions (HQ+Desc) achieved the best performance, though the advantage over HQ alone was modest (0.440 vs. 0.443, −0.7% for the single best model). Standardized laboratory photography (Nutrition5k) consistently yielded the poorest results across all models. Human Performance Analysis.

A board-certified nutritionist (SP) with more than ten years of clinical practice evaluated seven randomly selected food images from the HQ dataset using photographic visual estimation without any additional contextual information. Each dish required approximately one to two minutes of assessment. The nutritionist achieved an overall RMSLE of 0.176, with nutrient-specific accuracy varying markedly across macronutrients. Performance was highest for protein estimation (RMSLE: 0.098; MAE: 1.6 g), a result that can be reasonably attributed to the more identifiable and visually consistent features of protein-rich foods. In contrast, carbohydrate estimation yielded the largest relative errors (RMSLE: 0.362), reflecting well-known challenges such as heterogeneous carbohydrate density, visually ambiguous portioning, and non-visible sources of carbohydrates in sauces or preparation methods.

Comparison with model performance highlights substantial gaps. For proteins, the AI system exhibited markedly poorer accuracy, with errors more than fivefold larger in absolute terms (+581.3% MAE) and over sixfold larger on the RMSLE scale (+672.1%). Fat estimation likewise showed pronounced discrepancies, with AI errors exceeding human performance by approximately 152% (RMSLE) and 187% (MAE). Caloric estimation represented the only domain where AI systems approached human-level performance; the remaining accuracy gap was modest (+22.8% RMSLE; +32.3% MAE), suggesting that aggregate energy prediction is substantially more tractable for models than macronutrient decomposition. Collectively, these findings underscore that while AI approximates human ability in total caloric assessment, it remains considerably less reliable in distinguishing and quantifying individual macronutrient components from visual information alone.

Discussion

Principal findings

This systematic evaluation examined vision-language model performance across standardized laboratory photography (Nutrition5k dataset) and high-quality consumer images. Four primary findings emerged regarding the relative contributions of model architecture, image configuration, prompt design, and image quality to nutritional estimation accuracy.

Model architecture dominates all other factors

Model architecture accounted for 99.6% of performance variance, with between-model differences (up to 40% RMSLE variation) dwarfing within-model parameter effects by two orders of magnitude. Temperature (0.1%) and dataset seed (0.3%) had negligible effects. This dominance aligns with broader deep learning patterns where inductive biases, training data composition, and parameter scale override deployment-level interventions[12,13]. The best-performing model (GPT-5-chat, RMSLE 0.545) outperformed the worst by nearly 40%, indicating that model selection represents the single most consequential deployment decision—a finding consistent with recent vision-language model analyses[14,15].

Multi-image configurations provide no benefit

Multiple image angles conferred no performance advantage over single images (1-image RMSLE 0.627 vs. 5-images 0.623, $p = 0.182$), contradicting hypotheses that stereo or multi-view imaging would reduce portion estimation uncertainty[16]. This null result likely reflects two mechanisms. First, current VLMs lack explicit geometric reasoning for multi-view 3D reconstruction[3], treating additional images as redundant information rather than synthesizing volumetric estimates. Second, single high-quality overhead images may already saturate extractable visual information through plate coverage, shadow depth cues, and relative item sizing—visual features proven effective in previous dietary assessment systems[17]. Increased context length from multiple images may even dilute attention mechanisms[4], reducing focus on nutritionally-relevant features.

Prompt engineering shows no significant effects

Prompt engineering strategies showed no statistically significant performance differences after multiple comparison correction (baseline vs. 6 alternatives: all $p > 0.05$), despite chain-of-thought and few-shot learning demonstrating 10–50% improvements in text-based reasoning tasks[18,19]. This null result suggests fundamental differences in how VLMs process visual versus textual information[20]. Complex prompts may impose cognitive load without enhancing visual analysis[13], with models extracting nutritional information primarily through visual processing rather than linguistic guidance[9]. Simple and complex prompts appear functionally equivalent for vision tasks, indicating that deployment resources should focus on image quality and model selection rather than prompt optimization.

Image quality versus standardization

The most consequential practical finding emerged from comparing high-quality consumer photography against standardized laboratory imaging. High-quality consumer images significantly outperformed standardized Nutrition5k photography despite 56% smaller sample size (mean RMSLE: 0.548 vs. 0.616, $p = 0.020$, Cliff's $\delta = -0.58$, large effect, 12.8% improvement).

This result contradicts conventional assumptions that standardized conditions optimize AI performance. Higher resolution preserves fine texture details (meat marbling, surface char patterns) that inform density estimation. Moreover, natural lighting creates shadows revealing three-dimensional structure for volume assessment, whereas flattened overhead illumination eliminates depth cues. Contextual objects (i.e. plates and cutlery) could also help with VLM estimations. Additionally, consumer-style images better match the distribution of VLM training data[3], whereas standardized laboratory photography may induce a domain shift, degrading the utilization of learned features[12].

The large effect size ($\delta = -0.58$) across nine diverse architectures rules out model-specific artifacts, establishing image quality as a robust determinant of performance. For practical deployment, these results strongly indicate prioritizing ecological validity and resolution over dataset size or standardization—directly relevant for mobile applications where smartphone photography naturally provides high-quality consumer images[21].

Human-AI performance gap and current system limitations

Because the human comparator and the VLMs were evaluated on non-overlapping dish sets (7 high-quality consumer photographs for the nutritionist vs. 50 Nutrition5k photographs for the best VLM), the human–AI comparison is reported here as *indicative* rather than as a formal hypothesis test. With this caveat, the professional nutritionist maintained a substantial apparent performance advantage over every evaluated VLM (human RMSLE 0.176 vs. best VLM 0.443, a 152% relative gap, rising to 672% for protein estimation). Critically, this gap varies by nutrient: AI approaches human-level performance for total calories (+32% error), but struggles profoundly with macronutrient decomposition—particularly proteins (+672% RMSLE, +581% MAE). This pattern indicates VLMs can estimate food volume and apply approximate caloric density, but cannot replicate the compositional analysis, hidden ingredient identification (oils, sauces), and preparation method inference characterizing expert assessment. These findings align with recent evaluations showing systematic fat underestimation in VLM dietary assessment[5,6].

Textual ingredient descriptions (without quantities) produced statistically significant aggregate improvements (mean RMSLE: 0.548 → 0.516, $p = 0.039$, $\delta = 0.48$, 5.5% gain across 9 models), though with heterogeneous individual responses (−19.1% to +3.4%). This reveals architectural differences in multimodal fusion: Anthropic's Claude benefits substantially from text, while OpenAI's GPT-5-chat shows minimal sensitivity (−0.7%). The modest aggregate improvement (5.5%) compared to image quality effects (12.8%) confirms visual information dominates portion estimation, with text serving confirmatory roles for ingredient identification.

The five worst-performing dishes (mean RMSLE = 1.166) exhibited a consistent pattern: fat RMSLE averaged 1.855 (range 1.741–2.017), while protein and carbohydrate errors were comparatively moderate (0.995 and 0.706, respectively). This asymmetry indicates that VLMs can identify and quantify visible macronutrient sources (grains, vegetables, visible meat portions) but systematically underestimate invisible lipids from cooking methods (stir-frying, sautéing), fatty meat marbling, and emulsified sauces.

Conversely, the five best-performing dishes (mean RMSLE = 0.218) showed balanced nutrient estimation (RMSLE range 0.154–0.290 across all macronutrients), with fat estimation (mean RMSLE = 0.220) approaching the accuracy of carbohydrate estimation (0.257). This balanced performance confirms that when lipid content is visually apparent or minimal, current VLMs achieve near-human accuracy across all nutritional components.

This asymmetry indicates VLMs can quantify visible macronutrients (grains, vegetables, lean proteins) but systematically underestimate invisible lipids from cooking oils, fatty meat marbling, and emulsified sauces. Conversely, the five best-performing dishes (mean RMSLE = 0.218) showed balanced nutrient estimation (RMSLE range 0.154–0.290), confirming that when lipid content is visually apparent or minimal, current VLMs achieve near-human accuracy.

Performance benchmarking and study limitations

The best-performing model (GPT-5-chat, RMSLE 0.545, ~50% sMAPE) falls short of clinical standards. The Academy of Nutrition and Dietetics recommends ± 10% accuracy for dietary recalls[1], a threshold not approached by any model. These error rates align with recent VLM nutritional assessments reporting 20–50% underestimation[5,6] and likely reflect fundamental 2D image limitations: hidden ingredients (oils, sauces), preparation methods, and density variations remain occluded, information requiring 3D scanning or chemical analysis in ground-truth Nutrition5k[10].

Methodological limitations include modest sample sizes for human assessment ($N = 7$) and high-quality image comparisons ($N = 16$–22), limiting statistical power and precluding subgroup analyses. The human vs. AI comparison employed different dish sets, preventing formal hypothesis testing. Nutrition5k's controlled laboratory setting may not generalize to real-world smartphone images with variable lighting and backgrounds[21]. Moreover, pooled RMSLE gives nutrients the same weight. Different nutrients errors could have different relevance in different contexts, for example in sarcopenic patients' proteins errors should weigh more, while sugars should weigh more in diabetes. Future work should employ identical dish sets for human-AI comparison, larger sample sizes for high-quality photography experiments, and real-world smartphone capture conditions to establish ecological validity.

Priority future directions include: (1) direct human-versus-VLM comparisons on an identical, pre-registered dish set, using multiple board-certified nutritionists, to allow formal hypothesis testing of the human–AI gap reported here as indicative; (2) dedicated latency profiling against direct provider endpoints to inform edge and mobile deployment; (3) depth-augmented and 3D-reconstruction hybrid pipelines, in which Nutrition5k-style depth maps or smartphone LiDAR output are combined with VLM-based food recognition to decouple volume estimation from appearance-based density inference; and (4) empirical evaluation of hybrid VLM-plus-database pipelines, in which a VLM performs food identification and portion estimation and an external food-composition database provides per-item macronutrients, with the central empirical prediction that such pipelines should substantially reduce the compositional-error component of the human–AI gap while leaving the portion-estimation component largely unchanged.

With $n = 50$ paired observations, the Wilcoxon signed-rank tests used throughout have approximately 80% power to detect a Cliff's δ of 0.30 at $\alpha = 0.05$; the null results reported for multi-image configurations ($|\delta| \leq 0.21$) and prompt engineering strategies therefore reflect effects substantially smaller than those the sample can detect, rather than under-powered comparisons. Per-call inference latency was not recorded systematically and, in our setup, would in any case reflect the OpenRouter routing layer and upstream-provider queuing rather than intrinsic model characteristics.

Implications for practice and research

These findings yield five actionable deployment recommendations: (1) Prioritize model selection—architecture accounts for 99.6% of variance with up to 40% RMSLE differences between models. (2) Use single high-quality consumer-style photographs, natural lighting, contextual objects, and higher resolution significantly outperform standardized laboratory imaging ($p = 0.020$, 12.8% improvement), while multi-angle capture provides no benefit ($p = 0.182$). (3) Employ simple direct prompts—complex prompt engineering shows no significant effects (all $p > 0.05$) and may impose unnecessary cognitive load. (4) Collect optional ingredient descriptions—aggregate benefits are significant ($p = 0.039$, 5.5%) though model-dependent, with Claude architectures benefiting most (up to 19%) while GPT-5-chat shows minimal sensitivity. (5) Limit applications to consumer calorie tracking (± 50–100 kcal)—the 152% human-AI performance gap, particularly for protein estimation (+ 672%), precludes clinical macronutrient profiling requiring ± 10% precision.

Appropriate use cases include consumer dietary awareness applications, population-level trend analysis where individual errors average out, and screening tools for nutritional concerns. Hybrid VLM-plus-database approaches are a particularly promising direction.[22,15] A system that uses a VLM for food identification (and, optionally, for portion-size estimation) and then queries an external food-composition database (USDA FoodData Central, CIQUAL, or Open Food Facts) for per-item macronutrient values would replace the VLM's implicit compositional inference, which our results identify as the dominant source of error, with a calibrated and auditable estimate. We therefore expect such a hybrid to substantially reduce the compositional component of the human–AI performance gap (notably the 672% protein-specific gap) while leaving the portion-estimation component of the error largely unchanged, since portion estimation from 2D images remains ill-posed and compositionally complex dishes impose an additional irreducible ambiguity on any ingredient-based pipeline. We note that the evaluation reported here is deliberately restricted to single-call, single-model prompting, so that the model-architecture and deployment-axis variance is not confounded with additional system components; retrieval-augmented generation, tool-augmented prompting, and multi-agent pipelines are legitimate and complementary directions but address a different research question, namely the performance of composite systems rather than of the underlying VLM.

Conclusions

This systematic evaluation of 40 vision-language models reveals that model architecture selection dominates all other factors, accounting for 99.6% of performance variance with up to 40% RMSLE differences between models. In stark contrast, multi-image configurations, prompt engineering, and dataset standardization produce minimal or insignificant effects.

The most actionable finding contradicts standard practice: high-quality consumer smartphone photography significantly outperforms standardized laboratory imaging (12.8% improvement, large effect), demonstrating that ecological validity and visual richness trump-controlled standardization. Similarly, ingredient descriptions provide modest aggregate benefits (5.5% improvement) but with extreme model-dependent heterogeneity (−19% to +3%), indicating fundamental architectural differences in multimodal fusion.

However, professional nutritionists maintain a 152% performance advantage over the best AI systems, with the gap widening dramatically for macronutrient estimation (proteins: +672% RMSLE). Current VLMs estimate food volume and approximate caloric density adequately but cannot replicate expert compositional analysis or hidden ingredient detection, limiting applications to consumer calorie tracking (±50–100 kcal) rather than clinical macronutrient profiling, without hybrid human-AI validation.

Data availability

All analysis code, model predictions, original images, intermediate and final results, and experimental configurations are publicly available at https://github.com/lucavd/FoodPics_data_and_code. The Nutrition5k dataset is available at https://github.com/google-research-datasets/Nutrition5k under CC-BY 4.0 license.

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Author contributions

L.V. and C.L. concept, design, analysis l, and writing; S.P. concept, data collection; D.S. and M.J.B. analysis, writing, and review; G.L. concept and design; D.G. concept, design, and funding.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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